An Elementary Coset-Complex HDX with Ambient Group A_N

Abstract

We give a deliberately elementary construction of a bounded-degree family of 2-dimensional local spectral high-dimensional expanders whose ambient groups are alternating groups. The construction is a slight thickening of the rank-3 KMS triangle. The three local link bounds reduce to the facts that a nonzero linear polynomial has at most one root and a nonzero quadratic polynomial has at most two roots. The identification of the finite ambient group with an alternating group is also elementary: linear transvections plus one quadratic shear yield arbitrary one-variable shears, hence 3-transitivity and a literal 3-cycle.

1 The mother group

Fix

$$p = 11, \qquad R = \mathbf{F}_{11}[t].$$

Inside the group of polynomial automorphisms of R^3, with coordinates (x, y, z), define three finite subgroups

$$U_1 = \{a_s : (x, y, z) \mapsto (x + sy, y, z) : s \in \mathbf{F}_{11}\},$$
$$U_2 = \{b_s : (x, y, z) \mapsto (x, y + sz, z) : s \in \mathbf{F}_{11}\},$$
$$U_3 = \left\{c_{r,s} : (x, y, z) \mapsto (x, y, z + rtx + sx^2) : r, s \in \mathbf{F}_{11}\right\}.$$

Thus

$$U_1 \cong U_2 \cong (\mathbf{F}_{11}, +), \qquad U_3 \cong (\mathbf{F}_{11}^2, +).$$

Let

$$\Gamma = \langle U_1, U_2, U_3 \rangle.$$

Conceptually, this is the usual $(1, 1, 2)$ KMS triangle, except that the quadratic root group $z \leftarrow z+x^2$ has been enlarged by adjoining the commuting linear transvection $z \leftarrow z + tx$. This one extra direction removes the Frobenius obstruction in the finite quotients.

2 Finite quotients and the alternating group

Choose a monic irreducible polynomial

$$f(t) \in \mathbf{F}_{11}[t], \qquad \deg f = m \geq 2,$$

and write

$$K = \mathbf{F}_{11}[t]/(f) \cong \mathbf{F}_{11^m}, \qquad \theta = t \bmod f.$$

Let

$$\Omega = K^3 \setminus \{0\}, \qquad N = |\Omega| = 11^{3m} - 1.$$

Reducing the three groups modulo f gives permutations of Ω:

$$\begin{aligned} U_1 :&\quad x \leftarrow x + sy, \\ U_2 :&\quad y \leftarrow y + sz, \\ U_3 :&\quad z \leftarrow z + r\theta x + sx^2. \end{aligned}$$

Let G_f be the generated permutation group.

Theorem 1. *For every irreducible f of degree $m \geq 2$,*

$$G_f = A(\Omega) \cong A_{11^{3m}-1}.$$

The proof is broken into four elementary steps.

2.1 The linear elements generate $\mathrm{SL}_3(K)$

The group contains

$$e_{12}(1), \qquad e_{23}(1), \qquad e_{31}(\theta).$$

We repeatedly use the standard elementary-matrix commutator identity

$$[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$$

for distinct i, j, k. For example,

$$\begin{aligned} [e_{12}(1), e_{23}(1)] &= e_{13}(1), \\ [e_{23}(1), e_{31}(\theta)] &= e_{21}(\theta), \\ [e_{31}(\theta), e_{12}(1)] &= e_{32}(\theta). \end{aligned}$$

From $e_{31}(\theta^d)$ we obtain $e_{32}(\theta^d)$ and then

$$[e_{32}(\theta^d), e_{21}(\theta)] = e_{31}(\theta^{d+1}).$$

Inductively we obtain $e_{ij}(\theta^d)$ in every root position and for every $d \geq 0$. Since $1, \theta, \ldots, \theta^{m-1}$ span K over $\mathbf{F}_{11}$, additivity of the elementary root groups gives

$$e_{ij}(a) \in G_f \qquad (a \in K).$$

Hence the linear subgroup is

$$\mathrm{EL}_3(K) = \mathrm{SL}_3(K).$$

The group also contains the quadratic shear

$$\beta : (x, y, z) \mapsto (x, y, z + x^2).$$

Thus

$$G_f = \langle \mathrm{SL}_3(K), \beta \rangle.$$

2.2 All one-variable shears

For distinct coordinates i, j, write

$$T_{i\leftarrow j}(h): \quad x_i \longmapsto x_i + h(x_j),$$

fixing the other coordinates.

Lemma 2. *For every function $h : K \to K$ with $h(0) = 0$ and every $i \neq j$,*

$$T_{i\leftarrow j}(h) \in G_f.$$

Proof. We already have all linear shears, and by conjugating and scaling β we have all quadratic shears $x_i \leftarrow x_i + ax_j^2$.

Polarization gives a mixed product:

$$(x_j + x_k)^2 - x_j^2 - x_k^2 = 2x_jx_k,$$

so for distinct i, j, k we obtain a shear

$$x_i \leftarrow x_i + x_jx_k.$$

Suppose we already have

$$P: \ x_k \leftarrow x_k + x_j^d$$

and

$$Q: \ x_i \leftarrow x_i + x_jx_k.$$

Then, up to inversion depending on the commutator convention,

$$[P, Q]: \quad x_i \leftarrow x_i + x_j^{d+1}.$$

Starting from $d = 1$, induction yields every monomial shear and hence every polynomial shear with zero constant term.

Finally, every function $h : K \to K$ with $h(0) = 0$ is represented by a polynomial of degree at most $|K| - 1$ whose constant term is zero. This proves the claim. □

2.3 Three-transitivity

Lemma 3. *The action of G_f on Ω is 3-transitive.*

Proof. Take distinct nonzero points $v_1, v_2, v_3 \in K^3$. Choose a linear functional $\ell : K^3 \to K$ such that the three values

$$\ell(v_1), \ell(v_2), \ell(v_3)$$

are nonzero and pairwise distinct. The bad functionals lie in the six hyperplanes

$$\ell(v_i) = 0, \qquad \ell(v_i - v_j) = 0 \quad (i < j),$$

so their union has at most $6|K|^2 < |K|^3$ elements.

After applying an element of $\mathrm{SL}_3(K)$, write

$$v_i = (a_i, b_i, c_i)$$

with a_1, a_2, a_3 nonzero and pairwise distinct. By Lemma 2, using shears depending only on x, kill the second and third coordinates at these three points, obtaining $(a_i, 0, 0)$.

Now apply successively:

(1) $y \leftarrow y + x$, obtaining $(a_i, a_i, 0)$;

(2) choose $h(0) = 0$ with $h(a_i) = i - a_i$, and apply $x \leftarrow x + h(y)$;

(3) choose $r(0) = 0$ with $r(i) = -a_i$, and apply $y \leftarrow y + r(x)$.

The ordered triple becomes

$$(1, 0, 0), \quad (2, 0, 0), \quad (3, 0, 0).$$

Thus the action is 3-transitive. □

2.4 A literal 3-cycle

For $a \neq 0$, let $\delta_a : K \to K$ be the delta function

$$\delta_a(s) = \begin{cases} 1, & s = a, \\ 0, & s \neq a. \end{cases}$$

It satisfies $\delta_a(0) = 0$, so Lemma 2 makes the associated coordinate shears available.

We use the following elementary observation.

Lemma 4. *If two permutations σ, τ have supports intersecting in exactly one point, then their commutator $[\sigma, \tau]$ is a 3-cycle.*

Proof. Let w be the unique common support point. Away from the three points $w, \sigma(w), \tau(w)$, the two permutations commute. A direct check on these three points shows that the commutator cycles them. □

Define

$$A : x \leftarrow x + \delta_1(y), \qquad B : y \leftarrow y + \delta_1(x).$$

On every fixed z-slice the supports of A and B meet only at $(1, 1, z)$. Hence

$$D = [A, B]$$

is a product, over $z \in K$, of 3-cycles, and

$$\mathrm{supp}(D) = \{(1, 1, z), (2, 1, z), (1, 2, z) : z \in K\}.$$

Similarly define

$$C : y \leftarrow y + 10\delta_1(z), \qquad F : z \leftarrow z + \delta_3(y).$$

Then $E = [C, F]$ has support

$$\mathrm{supp}(E) = \{(x, 3, 1), (x, 2, 1), (x, 3, 2) : x \in K\}.$$

These two supports intersect in exactly one point, namely $(1, 2, 1)$. Therefore, by Lemma 4, a further commutator is a single 3-cycle.

Since G_f is 3-transitive, conjugating this 3-cycle gives every 3-cycle on Ω. Therefore

$$A(\Omega) \leq G_f.$$

Conversely, G_f is generated by elements of order 11, and every odd-order permutation is even. Hence

$$G_f \leq A(\Omega).$$

This proves Theorem 1.

Remark 5. The precise constants in the two displayed support families are unimportant. The only goal is to construct two products of disjoint 3-cycles whose supports meet in one point; the delta-shear mechanism makes this easy. This is the same elementary "supports meet once" idea that strips down the Gilman argument.

3 The three local pair groups

Let

$$P_{ij} = \langle U_i, U_j \rangle.$$

Put

$$W = \mathrm{span}_{\mathbf{F}_{11}}\{tx, x^2\}$$

and

$$V = \mathrm{span}_{\mathbf{F}_{11}}\{tx, ty, x^2, xy, y^2\}.$$

A direct commutator calculation gives normal forms

$$\begin{aligned}
P_{12} &= \{(x + ay + cz,\ y + bz,\ z) : a, b, c \in \mathbf{F}_{11}\}, \\
P_{23} &= \{(x,\ y + az + g(x),\ z + h(x)) : a \in \mathbf{F}_{11},\ g, h \in W\}, \\
P_{31} &= \{(x + ay,\ y,\ z + P(x, y)) : a \in \mathbf{F}_{11},\ P \in V\}.
\end{aligned}$$

In particular,

$$|P_{12}| = 11^3, \qquad |P_{23}| = 11^5, \qquad |P_{31}| = 11^6.$$

The same normal forms show

$$P_{12} \cap P_{23} = U_2, \qquad P_{23} \cap P_{31} = U_3, \qquad P_{31} \cap P_{12} = U_1,$$

and

$$P_{12} \cap P_{23} \cap P_{31} = \{1\}.$$

These statements remain true after reduction modulo any irreducible f of degree $m \geq 2$.

4 One Fourier lemma for all three link bounds

We isolate the only spectral input.

Lemma 6 (Polynomial-incidence lemma)**.** *Let N be a finite $\mathbf{F}_p$-vector space, and let*

$$V_s \leq N, \qquad s \in \mathbf{F}_p,$$

be subspaces spanning N. Form the bipartite incidence graph whose left vertices are the points $n \in N$, whose right vertices are the affine spaces

$$(s, n + V_s),$$

and where n is incident to $n + V_s$.

Suppose every nonzero linear functional $\lambda : N \to \mathbf{F}_p$ vanishes identically on V_s for at most d values of s. Then the second normalized eigenvalue of the bipartite graph is at most

$$\sqrt{d/p}.$$

Proof. Consider the two-step walk on the point side. Fourier characters diagonalize this walk. On the character indexed by λ, the eigenvalue is

$$\frac{1}{p}\left|\{s : \lambda|_{V_s} = 0\}\right| \le \frac{d}{p}.$$

The singular values of the one-step bipartite walk are the square roots of the eigenvalues of this two-step operator. □

We now apply the lemma three times.

4.1 The P_{12} link

Here the relevant subspaces are the lines

$$V_s = \mathrm{span}\{(s, 1)\} \le \mathbf{F}_{11}^2.$$

A nonzero linear functional vanishes on at most one such line, so

$$\lambda_2\big(\mathrm{CC}(P_{12}; U_1, U_2)\big) \le \frac{1}{\sqrt{11}}.$$

4.2 The P_{23} link

Here the normal abelian group is $W \oplus W$, and

$$V_s = \{(sw, w) : w \in W\}.$$

Again a nonzero functional annihilates at most one V_s, hence

$$\lambda_2\big(\mathrm{CC}(P_{23}; U_2, U_3)\big) \le \frac{1}{\sqrt{11}}.$$

4.3 The P_{31} link

Here the relevant subspaces are generated by the two polynomial directions

$$t(x + sy), \qquad (x + sy)^2.$$

Let L be a linear functional on

$$\mathrm{span}\{tx, ty, x^2, xy, y^2\}.$$

The condition $L|_{V_s} = 0$ becomes

$$L(tx) + sL(ty) = 0$$

and

$$L(x^2) + 2sL(xy) + s^2L(y^2) = 0.$$

If these equations hold for three distinct values of s, then both polynomials vanish identically, forcing $L = 0$. Thus a nonzero L can annihilate at most two of the subspaces V_s. Lemma 6 gives

$$\lambda_2\big(\mathrm{CC}(P_{31}; U_3, U_1)\big) \le \sqrt{\frac{2}{11}}.$$

Thus all three local calculations reduce to

$$\boxed{\text{a nonzero linear polynomial has at most one root}}$$

and

$$\boxed{\text{a nonzero quadratic polynomial has at most two roots}}.$$

5 The coset complex

Inside

$$G_f = A(\Omega),$$

define

$$H_1 = P_{23}, \qquad H_2 = P_{31}, \qquad H_3 = P_{12}.$$

Let

$$X_f = \mathrm{CC}(G_f; H_1, H_2, H_3).$$

Thus the vertices are the three types of cosets

$$G_f/H_1, \qquad G_f/H_2, \qquad G_f/H_3,$$

and the triangles are

$$\{gH_1, gH_2, gH_3\}, \qquad g \in G_f.$$

Because

$$H_1 \cap H_2 \cap H_3 = 1,$$

the group A_N acts sharply transitively on the triangles.

The three vertex links are exactly

$$\mathrm{CC}(P_{23}; U_3, U_2), \qquad \mathrm{CC}(P_{31}; U_1, U_3), \qquad \mathrm{CC}(P_{12}; U_2, U_1).$$

Hence every vertex link is connected and has

$$\lambda_2 \le \lambda := \sqrt{\frac{2}{11}} < \frac{1}{2}.$$

The elementary dimension-2 trickling-down theorem therefore gives

$$\lambda_2(X_f^{(1)}) \le \frac{\lambda}{1-\lambda} = \frac{\sqrt{2/11}}{1-\sqrt{2/11}} < 1.$$

Consequently the family $(X_f)_{\deg f \to \infty}$ is a bounded-degree family of 2-dimensional one-sided local spectral HDXs.

The degree is independent of m, since the three subgroups H_i have fixed orders $11^5, 11^6, 11^3$.

Theorem 7 (Elementary alternating-group HDX)**.** *For every irreducible $f \in \mathbf{F}_{11}[t]$ of degree $m \ge 2$, there is a 2-dimensional coset complex*

$$X_f = \mathrm{CC}\left(A_{11^{3m}-1}; H_1, H_2, H_3\right)$$

with the following properties:

(i) $A_{11^{3m}-1}$ acts sharply transitively on the triangles;

(ii) the degree is bounded independently of m;

(iii) the three vertex-link second eigenvalues are bounded by

$$\frac{1}{\sqrt{11}}, \quad \frac{1}{\sqrt{11}}, \quad \sqrt{\frac{2}{11}};$$

(iv) in particular, (X_f) is a uniform family of one-sided local spectral HDXs.

6 Property T as a byproduct

The same local estimates also give property T for the infinite mother group Γ. The corresponding pairwise subgroup Cayley gaps are at least

$$\frac{1}{2}\left(1-\sqrt{\frac{2}{11}}\right)>\frac{1}{4}.$$

Thus the three-subgroup criterion applies directly (with the natural weighting for the unequal subgroup sizes), yielding

$$\Gamma \text{ has property } (T).$$

For the HDX construction itself, however, property T is not needed separately: the local link bounds and the elementary trickling-down theorem already imply the global spectral expansion.

7 Summary

The complete construction is driven by the three finite root groups

$$\boxed{\begin{aligned}&U_1=\{x\leftarrow x+ay\},\\&U_2=\{y\leftarrow y+az\},\\&U_3=\{z\leftarrow z+atx+bx^2\}.\end{aligned}}$$

Their three pairwise coset graphs are controlled by linear, linear, and quadratic polynomial-incidence calculations, giving

$$\frac{1}{\sqrt{11}},\quad \frac{1}{\sqrt{11}},\quad \sqrt{\frac{2}{11}}<\frac{1}{2}.$$

After reduction modulo an irreducible polynomial f of degree m, the generated finite group is

$$A_{11^{3m}-1},$$

proved elementarily from arbitrary one-variable shears, 3-transitivity, and one literal 3-cycle. The resulting coset complexes are bounded-degree 2-dimensional local spectral HDXs with alternating ambient group.